# Data Free Training of Diffusion Models for Lattice Field Theories

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Sampling equilibrium field configurations remains a major computational bottleneck in lattice field theory, motivating the development of generative samplers that can produce independent configurations. Diffusion models are promising for this purpose, but their standard denoising score matching objective requires samples from the target distribution, seemingly reintroducing the original sampling problem. Here, we show that diffusion models can instead be trained from their own generated samples using the target score identity, provided access to the unnormalized probability density. Importance-weighted formulations of this identity yield a consistent target throughout training, while unweighted variants define a self-consistent fixed-point training scheme. We benchmark the different approaches in two-dimensional $\phi^4$ theory and identify effective choices that enable efficient training and sampling across the symmetric, critical, and broken phases. Ultimately, our data-free training approach achieves comparable performance to training on HMC reference data.

*Introduction.* Lattice field theory (LFT) is a cornerstone of the study of non-perturbative quantum field theories [1–4]. In lattice quantum chromodynamics in particular, first-principles simulations deliver quantities of direct phenomenological relevance such as quark masses, meson decay constants, and the strong coupling constant, all at controlled precision [5, 6]. These quantities are obtained from expectation values of observables, evaluated by sampling field configurations according to the Boltzmann distribution defined by the lattice action. The standard framework for this task is Markov chain Monte Carlo (MCMC), most commonly Hybrid Monte Carlo (HMC) [7]. However, as one approaches the critical region or continuum limit of the theory, MCMC suffers from *critical slowing down [8–10]* due to the long range correlation where the integrated autocorrelation time and the statistical error on some important observable grows at fixed chain length. Producing a fixed number of effectively independent configurations therefore becomes rapidly more expensive precisely in the regimes of physical interest.

Generative samplers offer a promising route to mitigating critical slowing down by directly producing statistically independent field configurations [11–16]. Substantial progress has been made with normalizing flows, from sampling scalar field theories [14, 17–27] to constructing samplers for lattice gauge theories [28–36] and developing approaches to address topological freezing [28, 30, 37, 38]. Despite these advances, the computational cost and architectural requirements of normalizing flows remain important limitations. Coupling-based flows impose architectural constraints to retain tractable density evaluation, while continuous normalizing flows offer greater architectural flexibility at the cost of expensive numerical integration and divergence evaluation during training.

Diffusion models provide an alternative to these flow-based approaches. Their success in machine learning applications [12, 39] has motivated growing interest in lattice field theory, with applications to scalar field theories [15, 40, 41] and lattice gauge theories [42–47]. A central obstacle to the practical use of diffusion models in lattice field theory is their reliance on representative training data or computationally expensive annealing procedures. Generating training ensembles with MCMC can reintroduce the original sampling bottleneck, while annealing-based approaches incur additional costs through sampling and correction steps across intermediate distributions. These limitations motivate a simple and computationally efficient data-free training approach that learns directly from the lattice action.

Recent work in the machine learning community has introduced data-free approaches to training diffusion samplers, including methods based on reverse Kullback–Leibler divergence over path measures [48–50], computationally expensive MCMC guided by the denoising score identity (DSI) [51, 52], and target score matching (TSM) [53], which relies on the target density and its gradient. However, their efficient application to lattice field theory remains challenging. For instance, importance-weighted methods, such as iDEM [54] are sensitive to poor overlap between proposal and target distributions, which can become increasingly severe as the lattice volume grows. Other approaches incur additional computational costs through trajectory-based training, as in SPS [55] or sampling and correction across intermediate distributions, as in PITA [56].

In this work, we establish TSM as a practical framework for data-free diffusion sampling in lattice field theory. TSM relates the diffusion score to the target-density gradient, allowing the lattice action to supply the learn-

ing signal. The model generates and periodically refreshes its own training configurations, as in iDEM [54] but learns through an unweighted, self-consistent objective that avoids importance-weight fluctuations during training, To stabilize this learning process, we combine a control-variate score target [57], which balances action-gradient and denoising information through a closed-form variance-minimizing coefficient, with a noise schedule normalized to fixed total variance. The resulting training procedure requires neither precomputed target ensembles nor a model-density evaluation.

We demonstrate the approach in two-dimensional scalar field theory across the symmetric, critical, and broken phases, obtaining asymptotically unbiased estimates of physical observables through importance reweighting. At criticality, the sampler achieves efficiencies comparable to the corresponding diffusion models trained on HMC ensembles, demonstrating that reference data are not required to achieve this sampling performance. The unweighted training scheme enables stable learning and efficient sampling at lattice volumes where importance-weighted training becomes unstable. These results establish a practical route to learning equilibrium lattice distributions directly from the action, retaining accurate physical predictions while avoiding the costs of generating MCMC training ensembles and performing additional annealing procedures.

*Method.* We consider a real scalar field on a $D$-dimensional hypercubic lattice $\Lambda_L = \{1, \dots, L\}^D$, with a field variable $\phi_{\mathbf{x}} \in \mathbb{R}$ at each site $\mathbf{x} \in \Lambda_L$. A field configuration is the collection $\boldsymbol{\phi} = \{\phi_{\mathbf{x}}\}_{\mathbf{x}\in\Lambda_L} \in \mathbb{R}^{|\Lambda_L|}$. The action over field configurations is defined as:

$$S[\boldsymbol{\phi}] = \sum_{\mathbf{x}} \big[ -2\kappa \sum_{\mu} \phi_{\mathbf{x}}\phi_{\mathbf{x}+\hat{\boldsymbol{\mu}}} + (1-2\lambda)\phi_{\mathbf{x}}^2 + \lambda\phi_{\mathbf{x}}^4 \big]. \tag{1}$$

Our goal is to sample field configurations from the Boltzmann distribution:

$$p(\boldsymbol{\phi}) = \frac{1}{Z} e^{-S[\boldsymbol{\phi}]}, \text{ where } Z = \int \mathcal{D}\boldsymbol{\phi}\, e^{-S[\boldsymbol{\phi}]}, \mathcal{D}\boldsymbol{\phi} = \prod_{\mathbf{x}} d\phi_{\mathbf{x}}. \tag{2}$$

We construct a diffusion sampler by learning to reverse a process that transforms field configurations into Gaussian noise. The reverse dynamics require the score, $\nabla \log p_t$, of the marginal distribution at each diffusion time, which we approximate with a neural network. The diffusion time runs over $t \in [0, 1]$ with

$$\begin{aligned} t = 0: &\ \ \boldsymbol{\phi}(0) \sim p \propto e^{-S} \quad \text{(target distribution)}, \\ t = 1: &\ \ \boldsymbol{\phi}(1) \sim p_1 = \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad \text{(Gaussian prior)}. \end{aligned} \tag{3}$$

Noising therefore runs *forward* in $t$, from a field configuration at $t = 0$ to the prior at $t = 1$, and generation runs backward, from $t = 1$ down to $t = 0$. The noising process is the linear Itô SDE:

$$d\boldsymbol{\phi}(t) = f(t)\,\boldsymbol{\phi}(t)\,dt + g(t)\,d\mathbf{w}(t), \qquad \boldsymbol{\phi}(0) \sim p, \tag{4}$$

where $f(t)$ is an affine drift, $g(t)$ is the diffusion coefficient, and $\mathbf{w}$ is a Wiener process. Conditioned on its starting point $\boldsymbol{\phi}(0)$, this process has a closed-form Gaussian *transition kernel*:

$$\begin{aligned} p\big(\boldsymbol{\phi}(t) \,\big|\, \boldsymbol{\phi}(0)\big) &= \mathcal{N}\big(\boldsymbol{\phi}(t);\, a(t)\,\boldsymbol{\phi}(0),\, b^2(t)\,\mathbf{I}\big), \\ a(0) = 1,\ b(0) = 0,\ a(1) &\to 0,\ b(1) \to 1, \end{aligned} \tag{5}$$

which we can sample directly, i.e. without simulation, during training.

Differentiating the perturbation kernel (5) at a fixed endpoint gives the denoising score identity (DSI):

$$\begin{aligned} \nabla \log p_t &= \mathbb{E}\big[\mathbf{s}_{\mathrm{DSI}} \,\big|\, \boldsymbol{\phi}(t)\big], \text{ where} \\ \mathbf{s}_{\mathrm{DSI}} &:= \nabla_{\boldsymbol{\phi}(t)} \log p\big(\boldsymbol{\phi}(t) \,\big|\, \boldsymbol{\phi}(0)\big) \\ &= -\frac{\boldsymbol{\phi}(t) - a(t)\boldsymbol{\phi}(0)}{b^2(t)} = -\frac{\boldsymbol{\epsilon}}{b(t)}. \end{aligned} \tag{6}$$

Here $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ is the noise entering Eq. (5). This is the standard denoising score matching (DSM) target. Forming it is cheap, since it needs only the noise $\boldsymbol{\epsilon}$ already drawn and the configuration $\boldsymbol{\phi}(0)$, and it never touches the action. However, it has two drawbacks: First, it requires equilibrium sample from the target distribution $\boldsymbol{\phi}(0) \sim p$, which is unavailable in the data-free setting. Second, as the added noise vanishes near the physical endpoint, $t \to 0$, the factor $1/b(t)$ amplifies the Gaussian noise in the regression target. Its variance therefore grows as $1/b(t)^2$, making the training signal increasingly noisy even though the underlying physical score remains finite.

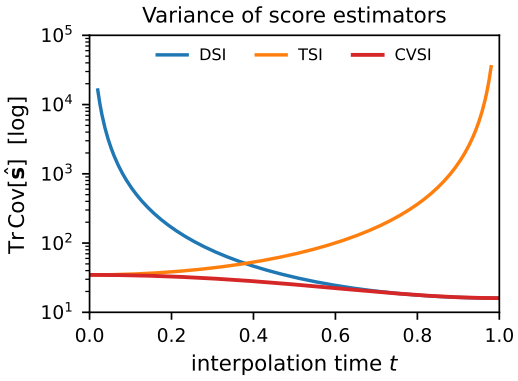



FIG. 1. Score estimator variance vs. diffusion time $t$ ($L = 4$, symmetric). While DSI and TSI diverge at either end of the time interval, CVSI variance stays low.

As [53] show, the marginal score can be equally rewritten as

$$\begin{aligned} \nabla \log p_t &= \mathbb{E}\big[\mathbf{s}_{\mathrm{TSI}} \,\big|\, \boldsymbol{\phi}(t)\big] \text{ where} \\ \mathbf{s}_{\mathrm{TSI}} &:= \frac{1}{a(t)} \nabla_{\boldsymbol{\phi}(0)} \log p\big(\boldsymbol{\phi}(0)\big) = -\frac{\nabla S\big[\boldsymbol{\phi}(0)\big]}{a(t)}, \end{aligned} \tag{7}$$

yielding the target score identity (TSI). This TSM target involves the action gradient, which encodes information about the target distribution and thus enables data-free training using self-generated field configurations. Its drawback, however, is that the $1/a(t)$ pre-factor diverges at the prior end $t \to 1$, where $a(t) \to 0$.

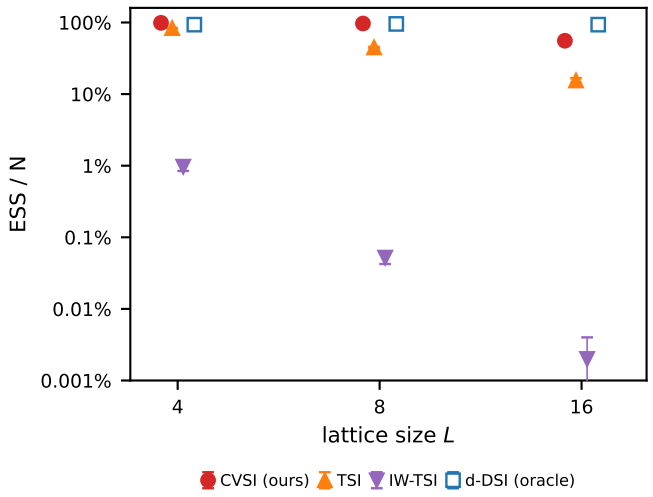



FIG. 2. ESS comparison of our data-free CVSI training (red circle), TSI (orange triangle) and IW-TSI (purple inverted triangle), as well as the oracle data-driven DSI (blue square), at criticality. Error bars show $1\sigma$ confidence intervals.

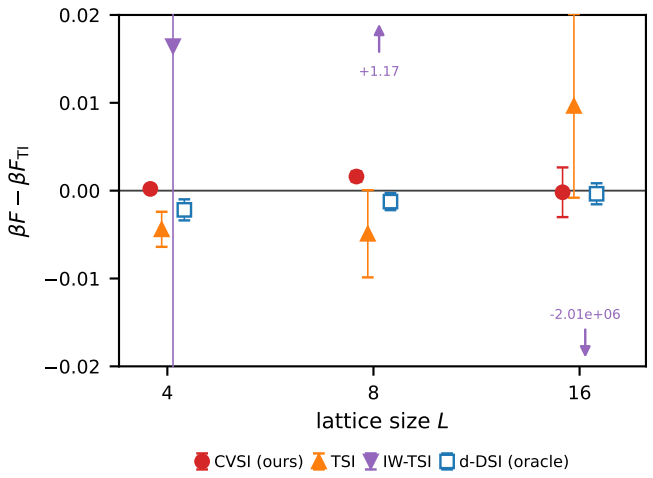



FIG. 3. Free-energy estimation error $\beta F - \beta F_{\mathrm{TI}}$ at criticality.

DSI (6) and TSI (7) provide two unbiased estimators for the *same* marginal score, which diverge at opposite ends of the schedule, as Fig. 1 shows. The control variate score identity (CVSI) [57] uses the optimal combination:

$$\hat{\mathbf{s}}_{\mathrm{CVSI}} = \big(1 - c^*(t)\big)\,\mathbf{s}_{\mathrm{TSI}} + c^*(t)\,\mathbf{s}_{\mathrm{DSI}},\ \ c^*(t) \in [0,1], \quad (8)$$

where the coefficient $c^*(t)$ is the minimizer of the variance of the blended target. While the general coefficient $c^*$ [57] is state- and time-dependent, for a Gaussian target with per-site variance $\sigma^2_{\mathrm{data}}$ and the kernel (5), this minimizer reduces to the closed form,

$$c^*(t) = \frac{b^2(t)}{b^2(t) + a^2(t)\,\sigma^2_{\mathrm{data}}} = \frac{1}{1 + \gamma^2(t)\,\sigma^2_{\mathrm{data}}}, \quad (9)$$

where $\gamma^2(t) = a^2(t)/b^2(t)$ is the signal-to-noise ratio (Supplemental Material, Eq. (8)).

As shown in Fig. 1, the CVSI variance remains finite throughout the diffusion process $t \in [0,1]$. At the field end $t \to 0$, $c^*(t) \to 0$ as $b(t) \to 0$, suppressing the DSI term whose $1/b(t)$ prefactor diverges, while at the prior end $t \to 1$, $c^*(t) \to 1$ as $a(t) \to 0$, suppressing the TSI term whose $1/a(t)$ prefactor diverges. This can also be seen from the explicit expression of the CVSI target (13) in the End Matter, obtained by substituting Eqs. (6), (7), and (9) into Eq.(8). For any $\sigma^2_{\mathrm{data}} > 0$, the CVSI target remains bounded throughout the entire schedule, requiring only one action-gradient computation per training point.

Following [58], we adopt a mixed noise schedule combining variance-exploding (VE) and variance-preserving (VP) schedules. Specifically, we match the (inverse) signal-to-noise ratio of the VE schedule, i.e. $b(t)/a(t) = \sigma(t) = \gamma^{-1}(t)$, while matching the total variance of the VP schedule, i.e. $\tau^2(t) = a^2(t) + b^2(t) = 1$. This gives the transition kernel (5) with

$$a(t) = \frac{1}{\sqrt{1+\sigma^2(t)}}, \qquad b(t) = \frac{\sigma(t)}{\sqrt{1+\sigma^2(t)}},$$
$$\sigma(t) := \sigma_{\min}\Big(\frac{\sigma_{\max}}{\sigma_{\min}}\Big)^t. \quad (10)$$

The noise level increases from $\sigma(0) = \sigma_{\min}$ at the field end to $\sigma(1) = \sigma_{\max}$ at the prior end, as required by our time convention (Supplemental Material, Eq. (4)), so $a(0) \to 1$ and $a(1) \to 0$. Bounding the total variance at $\tau^2 = 1$ makes the prior $p_1 = \mathcal{N}(\mathbf{0}, \mathbf{I})$ independent of $\sigma_{\max}$ and puts $a$ and $b$ on a common scale, so that the blend coefficient of the target is set by the single scalar $\sigma^2_{\mathrm{data}}$. We use $\sigma_{\min} = 10^{-2}$ throughout and treat $\sigma_{\max}$ as the one schedule hyperparameter, sweeping it below.

We train a score model $\mathbf{s}_{\boldsymbol{\theta}}(\boldsymbol{\phi}(t), t) \approx \nabla_{\boldsymbol{\phi}(t)} \log p_t(\boldsymbol{\phi}(t))$ on pairs drawn from the kernel (5), with $\boldsymbol{\phi}(0)$ drawn from the buffer. The only access we assume to the target is pointwise evaluation of $S[\boldsymbol{\phi}]$ and $\nabla S[\boldsymbol{\phi}]$. Both score identities are available on such a pair, and we blend them as in Eq. (8). In the data-free setting there is no initial dataset to read $\sigma^2_{\mathrm{data}}$ from. We therefore hold it at $\sigma^2_{\mathrm{data}} = 1$ for the first $R_{\mathrm{warm}}$ rounds and thereafter re-estimate it at each buffer refresh using the model's own configurations:

$$\sigma^2_{\mathrm{data}} \leftarrow \mathrm{clip}_{[\sigma^2_-, \sigma^2_+]}\Big(\mathrm{Var}\big[\mathcal{B}\big]\Big), \quad (11)$$

where the variance is computed over all sites across all $N_{\mathrm{gen}}$ buffer configurations. We use $R_{\mathrm{warm}} = 100$ and $[\sigma^2_-, \sigma^2_+] = [0.25, 10]$. We apply these safeguards during the initial rounds because the round-0 buffer, generated by an untrained sampler, yields variance estimates of order $10^3$–$10^5$, whereas the true variance is of order one. Updating $\sigma^2_{\mathrm{data}}$ adds negligible cost, since we produce the buffer samples anyway and the variance is fast to compute.

Training alternates two loops. In the outer loop we draw $N_{\mathrm{gen}}$ configurations $\boldsymbol{\phi}(0)$ by integrating the reverse SDE from the prior at $t = 1$ down to $t = 0$ with the

TABLE I. Free energy per site $f = \beta F/|\Lambda_L|$ and TI reference $f_{\rm TI}$. Errors in parentheses are $1\sigma$ on the last digits. The signed score $z = (f - f_{\rm TI})/\sqrt{\sigma_f^2 + \sigma_{f_{\rm TI}}^2}$ uses independent sampler and TI standard errors. Symmetric pools use $N = 50,000$. Critical pools use $N = 100,000$. Broken pools use $N = 50,000$.

| $L$ | phase | $\beta F/\lvert\Lambda_L\rvert$ | $f_{\rm TI}$ | ESS (%) | $z$ |
|---|---|---|---|---|---|
| 4 | symmetric | −0.687128(13) | −0.687127(39) | 99.8 | −0.03 |
| | critical | −0.759598(22) | −0.759610(54) | 98.8 | +0.21 |
| | broken | −0.828053(29) | −0.827985(67) | 98.9 | −0.93 |
| 8 | symmetric | −0.670360(7) | −0.670380(17) | 99 | +1.09 |
| | critical | −0.721223(9) | −0.721248(25) | 96.7 | +0.94 |
| | broken | −0.787032(11) | −0.787052(35) | 97.4 | +0.56 |
| 16 | symmetric | −0.669506(3) | −0.669511(9) | 96.1 | +0.46 |
| | critical | −0.713112(11) | −0.713111(13) | 55.6 | −0.04 |
| | broken | −0.778631(16) | −0.778651(20) | 55.8 | +0.75 |

current model. These configurations replace the replay buffer $\mathcal{B}$. We also store their action gradients and re-estimate $\sigma^2_{\rm data}$. This is the only place where trajectories are simulated. In the inner loop we draw $\boldsymbol{\phi}(0)$ from $\mathcal{B}$, a time $t$ and fresh noise $\boldsymbol{\epsilon}$, form $\boldsymbol{\phi}(t)$ in one shot with Eq. (5), build the CVSI target (8), and take a gradient step. No trajectory is simulated in the inner loop and no gradient passes through an SDE solve. After $E$ steps the buffer is refreshed and the round repeats. We keep an exponential moving average of the parameters and use it for generation and evaluation. The procedure is summarised as Algorithm 1 in the End Matter.

*Results.* We first compare the performance of different data-free score-estimator targets and a data-driven oracle target at criticality ($\kappa = 0.2705$) on $L \in \{4, 8, 16\}$ lattices. The quartic coupling constant $\lambda = 0.022$ is held fixed throughout all experiments. Figures 2 and 3 show ESS and the the free-energy estimation error $\beta F - \beta F_{\rm TI}$, respectively, where the thermodynamic integration (TI) serves as the reference. Our best choice, CVSI, ourperforms IW-TSI and TSI, which correspond to the direct application of iDEM [54] and adjoint matching [59], respectively, and is comparable to the data-driven DSI (d-DSI)—the standard diffusion training on the HMC samples.

We also evaluated our CVSI training in symmetric ($\kappa = 0.25$) and broken ($\kappa = 0.28$) phases. The results are reported in Table I, whose right-most column gives the signed compatibility $z$-score $z = (f - f_{\rm TI})/\sqrt{\sigma_f^2 + \sigma_{f_{\rm TI}}^2}$. The estimate is considered statistically compatible with the TI reference if $|z| \lesssim 2$–$3$. In all cases, CVSI training yields unbiased free-energy estimates across lattice sizes and phases.

Figure 4 shows estimation errors for other observables at criticality, using HMC sampling as the reference. These include the mean absolute magnetization $\langle|m|\rangle$, the susceptibility of the absolute magnetization $\chi_{|M|}$, the magnetic susceptibility $\chi_M$, and the fourth-order Binder cumulant $U_4$. For all observables and lattice sizes, the estimates are statistically compatible with the HMC reference and d-DSI oracle, suggesting that data-free training does not suffer from mode-dropping. Table I in the Supplemental Material summarizes the results.

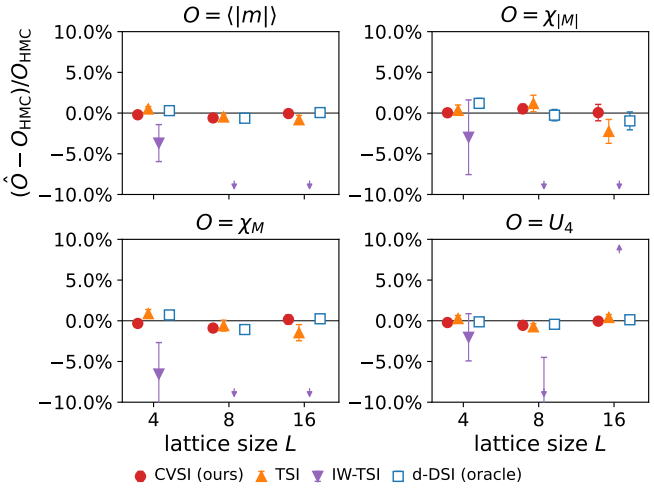



FIG. 4. Estimation errors for the mean absolute magnetization $\langle|m|\rangle$, the susceptibility of the absolute magnetization $\chi_{|M|}$, the magnetic susceptibility $\chi_M$, and the fourth-order Binder cumulant $U_4$.

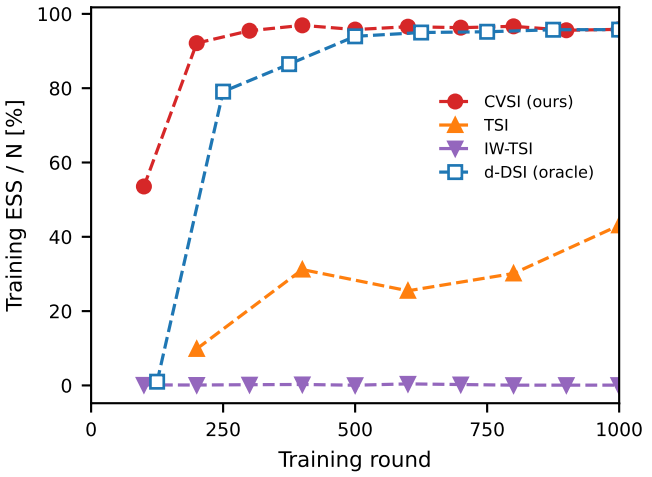



FIG. 5. Training dynamics of the ESS for $N = 5000$ samples.

As seen in Figures 2, importance weighting tends to degrade performance, while CVSI mitigates the main weakness of TSI: the divergence of the variance as $t \to 1$. Our ablation study in the End Matter confirms that these trends persist across different combinations of score estimators, data-free and data-driven training, and the use or omission of importance weighting.

Lastly, Fig. 5 shows the ESS as a function of training steps for the critical phase at $L = 8$. We observe that IW-TSI struggles to learn the target distribution

due to its high variance. CVSI accelerates the TSI training, and converges as fast as d-DSI without using HMC samples. These results again highlight the importance of controlling target variance for stable and efficient data-free training.

*Conclusion.* We have demonstrated that diffusion models can be trained for equilibrium sampling in lattice field theory without requiring reference configurations from the target distribution or a costly annealing procedure. By exploiting the target score identity (TSI), our approach enables data-free training using self-generated configurations. Our benchmarks in two-dimensional $\phi^4$ theory show that blending denoising score identity (DSI) with optimal balance—yielding the control variate score identity (CVSI)—significantly improves sampler performance, and importance weighting—which fixes the expectation of the target score across training steps—is harmful for stability due to its large variance. Our suitable choice of training targets enable efficient training and sampling across the symmetric, critical, and broken phases. Remarkably, the resulting samplers achieve performance comparable to models trained on HMC reference data, demonstrating that access to target-distributed training samples is not a prerequisite for effective diffusion-based sampling. This opens a promising direction for developing generative samplers for lattice field theories where generating reference configurations is itself computationally expensive.

Several promising directions remain for future work. Since our data-free training approach is orthogonal to annealing-based methods, combining the two could further improve scalability and enable sampling on larger lattices. Its compatibility with multilevel sampling strategies also warrants investigation. Beyond scalar field theories, extending the approach to lattice gauge theories and larger lattice volumes will be important for establishing its broader applicability.

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# End Matter

## I. LATTICE SIZES, PHASES, NETWORK ARCHITECTURE, AND TRAINING ALGORITHM

All experiments use the $D = 2$ theory on periodic square lattices $\Lambda_L = \{1, \dots, L\}^2$ at fixed $\lambda = 0.022$. We consider three sizes and, through the hopping parameter $\kappa$, three phases:

$$L \in \{4, 8, 16\} \quad (|\Lambda_L| = L^2 \in \{16, 64, 256\}),$$
$$\kappa \in \{\underbrace{0.25}_{\text{sym.}}, \underbrace{0.2705}_{\text{crit.}}, \underbrace{0.28}_{\text{broken}}\}.$$

For each of the nine $(L, \kappa)$ ensembles an HMC reference of $10^5$ configurations is used for the reference.

*UNet.* We parameterize the score with a time-conditioned convolutional UNet [12, 15, 40, 41, 45, 60], using residual blocks, attention, and skip connections. Architecture details and size-dependent settings are given in Supplemental Material, Sec. XI.

*Circular padding.* Spatial convolutions wrap across opposite lattice edges, implementing periodic boundary conditions.

$\mathbb{Z}_2$ *symmetry.* The score is made odd under field inversion by the model-level symmetrization

$$\mathbf{s}_{\boldsymbol{\theta}}^{\mathbb{Z}_2}(\boldsymbol{\phi}, t) = \tfrac{1}{2}[\mathbf{s}_{\boldsymbol{\theta}}(\boldsymbol{\phi}, t) - \mathbf{s}_{\boldsymbol{\theta}}(-\boldsymbol{\phi}, t)]. \tag{12}$$

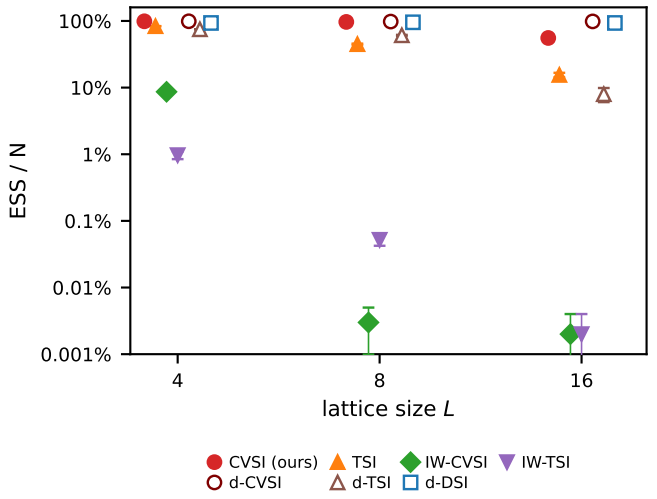



FIG. 6. ESS comparison of data-free and data-driven training.

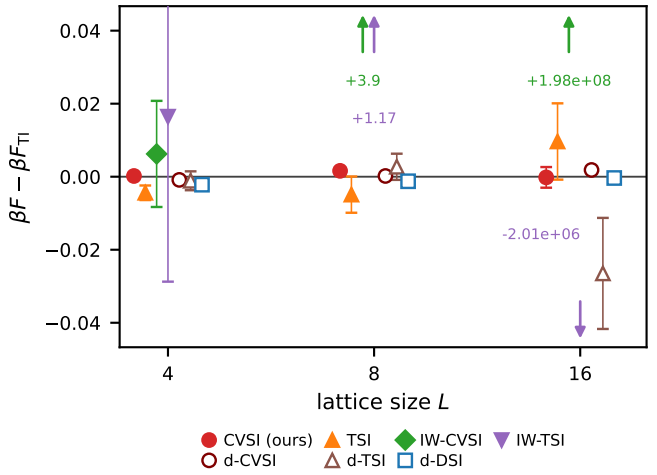



FIG. 7. Free-energy estimator errors $\beta F - \beta F_{\text{TI}}$.

**Algorithm 1:** Data-free training via fixed-point iteration.

**Input:** $S$, $(a, b, w)$, $R$, $E$, $N_{\text{gen}}$, $\eta$, $\alpha_{\text{EMA}}$, $\Omega$, $R_{\text{warm}}$, $[\sigma_-^2, \sigma_+^2]$
Initialize $\boldsymbol{\theta}$, $\bar{\boldsymbol{\theta}} \leftarrow \boldsymbol{\theta}$, $\sigma_{\text{data}}^2 \leftarrow 1$.
**for** $i = 1, \dots, R$ **do**
  Generate $N_{\text{gen}}$ fields with $\mathbf{s}_{\bar{\boldsymbol{\theta}}}$ and replace $\mathcal{B}$.
  **if** $i > R_{\text{warm}}$ **then**
    Update $\sigma_{\text{data}}^2$ using Eq. (11).
  **end if**
  **for** $e = 1, \dots, E$ **do**
    Draw $\boldsymbol{\phi}(0) \sim \mathcal{B}$, $t \sim \mathcal{U}[0.01, 0.99]$,
    $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$; form $\boldsymbol{\phi}(t) = a(t)\boldsymbol{\phi}(0) + b(t)\boldsymbol{\epsilon}$.
    Compute $c^*(t)$ and $\hat{\mathbf{s}}$ from Eqs. (9), (14).
    $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \nabla_{\boldsymbol{\theta}} \left[ w(t) \| \mathbf{s}_{\boldsymbol{\theta}}(\boldsymbol{\phi}(t), t) - \hat{\mathbf{s}} \|^2 \right]$
    $\bar{\boldsymbol{\theta}} \leftarrow \alpha_{\text{EMA}} \bar{\boldsymbol{\theta}} + (1 - \alpha_{\text{EMA}}) \boldsymbol{\theta}$
  **end for**
**end for**
**Output:** $\mathbf{s}_{\bar{\boldsymbol{\theta}}}$.

## II. THE BLENDED CVSI TARGET AND CLIPPING

Substituting the score identities and the optimal coefficient of Eq. (9) gives

$$\hat{\mathbf{s}}_{\text{CVSI}}(\boldsymbol{\phi}(0), \boldsymbol{\epsilon}, t) = -\frac{a(t)\, \sigma_{\text{data}}^2\, \nabla S[\boldsymbol{\phi}(0)] + b(t)\, \boldsymbol{\epsilon}}{b^2(t) + a^2(t)\, \sigma_{\text{data}}^2}. \tag{13}$$

For $\sigma_{\text{data}}^2 > 0$, the blend removes the individual $1/a(t)$ and $1/b(t)$ singularities. For stability, each identity is norm-clipped before blending:

$$\hat{\mathbf{s}}_{\text{CVSI}} = \big(1 - c^*(t)\big)\, \text{clip}_\Omega\big(\mathbf{s}_{\text{TSI}}\big) + c^*(t)\, \text{clip}_\Omega\big(\mathbf{s}_{\text{DSI}}\big),$$
$$\text{clip}_\Omega(\mathbf{s}) = \mathbf{s} \cdot \min\Big(\frac{\Omega}{\|\mathbf{s}\|}, 1\Big), \tag{14}$$

We use $\Omega = 70$. Clipping limits the influence of large gradients from early self-generated buffers.

Algorithm 1 summarizes the training loop.

## III. COMPARISON TO OTHER DATA-DRIVEN ESTIMATORS

Figs. 6 and 7 show the comparison of our data-free training to data-driven training in terms of ESS and

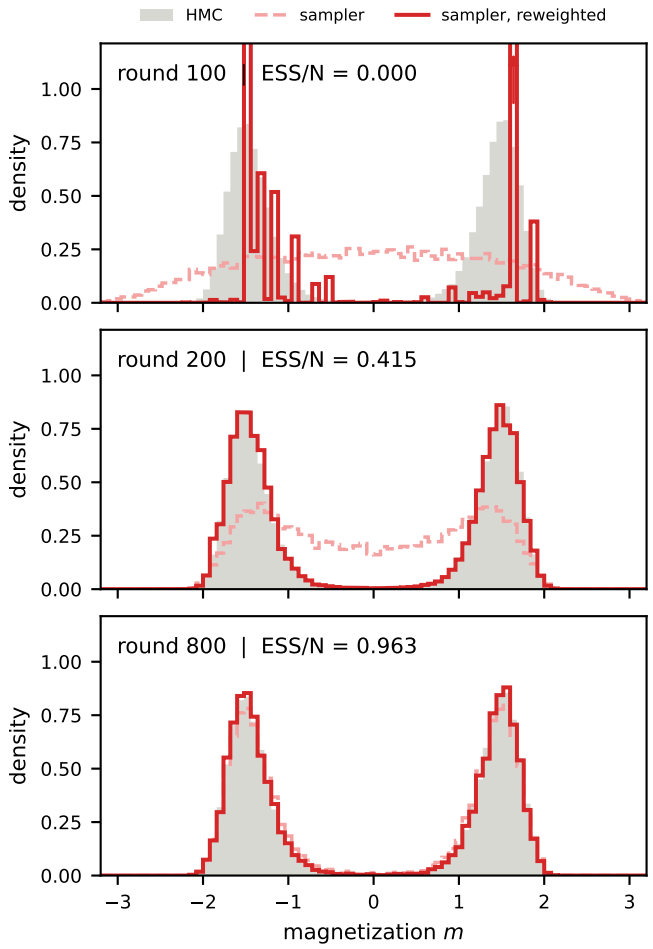



FIG. 8. Magnetization at $L = 8$, broken phase, over the self-learning loop. Raw samples (light red) and reweighted (red) approach the HMC reference (grey).

free-energy estimator error. We observed general trends: CVSI improves TSI and DSI, and IW substantially degrades performance.

## IV. OBSERVABLES AND EVALUATION METRICS

The magnetization and its absolute value are

$$m[\boldsymbol{\phi}] = \frac{1}{|\Lambda_L|}\sum_{\mathbf{x}\in\Lambda_L}\phi_{\mathbf{x}}, \qquad |m|[\boldsymbol{\phi}] = \Big|\frac{1}{|\Lambda_L|}\sum_{\mathbf{x}\in\Lambda_L}\phi_{\mathbf{x}}\Big|, \tag{15}$$

where $|\Lambda_L| = L^D$. The full distribution tests both relative mode populations and mode shapes; $\langle m\rangle \simeq 0$ alone does not establish agreement with HMC.

Figure 8 shows the evolution of the raw and reweighted magnetization distributions at $L = 8$ in the broken phase over the self-learning loop.

*ESS & Free Energy* We compute the work for each sample, i.e.

$$W[\boldsymbol{\phi}] = S[\boldsymbol{\phi}] + \log q(\boldsymbol{\phi}), \tag{16}$$

where $q$ is the density of the learned sampler, from which the ESS and free energy are computed as

$$\frac{\mathrm{ESS}}{N} = \frac{\langle e^{-W}\rangle_q^2}{\langle e^{-2W}\rangle_q}, \qquad \beta F = -\log\langle e^{-W}\rangle_q. \tag{17}$$

The associated self-normalized importance weights are $\omega_i = \frac{e^{-W_i}}{\sum_i e^{-W_i}}$.